

Automatic Construction System for Curriculum Knowledge Graphs Based on Large Language Models LLM-Based Curriculum Map Construction

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Abstract

This study designs and implements an automated knowledge graph construction system for teaching syllabus based on large language models, aiming to address the low efficiency and difficulty in updating associated with traditional manual construction methods. Through a multi-stage processing workflow involving entity recognition, relation extraction, and context-aware semantic fusion, the system automatically extracts entity relationships such as courses, knowledge points, and textbooks from teaching syllabus documents and constructs structured knowledge graphs. Experiments validated the system using syllabus from three courses: Data Structures, Object-Oriented Programming, and Compiler Principles. Average F1 scores for entity and relationship extraction exceeded 0.7, demonstrating high accuracy and practicality in knowledge extraction and relationship construction. The system holds promising applications in intelligent educational resource management and personalized learning path recommendations.

CCS Concepts

• **Information systems** → Information storage systems; Storage architectures.

Keywords

Knowledge Graph, Large Language Models, Intelligent Education

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1 Introduction

As the education sector increasingly embraces digitalization, structured data frameworks like knowledge graphs have garnered significant attention in educational applications. Traditional methods for managing teaching syllabus and organizing knowledge often rely on manual curation, resulting in inefficiency and difficulty in updates. A key research focus in educational technology is how to move beyond manual approaches by leveraging advanced natural language processing techniques to extract and structure knowledge points and their relationships from syllabus-type documents, thereby constructing high-quality knowledge graphs.

A knowledge graph is a semantic network that describes concepts and their relationships in a graph structure, effectively representing connections between pieces of knowledge. In educational settings, knowledge graphs can clearly visualize instructional structures. This facilitates personalized learning and provides technical support for implementing intelligent recommendations and teaching quality assessments [1] [8] [9]. However, traditional knowledge graph construction method rely heavily on domain experts, resulting in high costs, low efficiency, and difficulty adapting to the constant evolution and updates of teaching content.

In recent years, breakthroughs in deep learning-based natural language processing have opened new avenues for automated knowledge extraction in the knowledge and the relationship between them. Large-scale pre-trained language models demonstrate strong text comprehension and information extraction capabilities, providing the technical foundation for automatically identifying entities and relationships within unstructured syllabus documents [2][10]. This makes constructing high-quality knowledge graphs feasible.

This study designed and implemented a deep learning-based intelligent system addressing the critical challenge of automated syllabus knowledge graph construction. The system employs a multi-stage processing workflow: first, pre-trained language models identify entities; subsequently, relationship extraction algorithms recognize entity connections like prerequisite and inclusion relationships^[3]; finally, domain-specific knowledge is integrated to construct structured knowledge graphs. The system supports input from multiple document formats and holds promise as a new

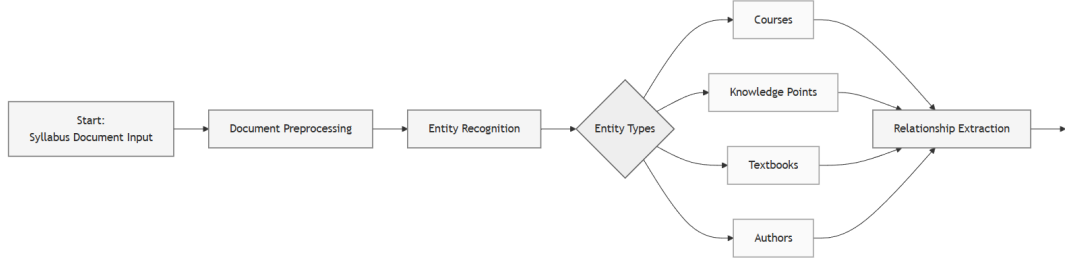


Figure 1: Flowchart 1, Process Flow for Automatically Extracting Structured Knowledge from Syllabus Documents

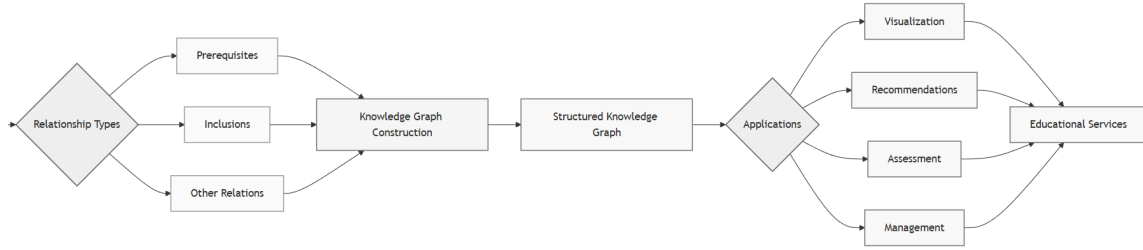


Figure 2: Flowchart 2, Constructing a Structured Knowledge Graph

knowledge organization tool for educational administrators and teachers.

To validate the system’s capabilities, this study constructed multiple computer science course syllabus data files, including courses such as Data Structures, Compiler Principles, and Object-Oriented Programming.

The contributions of this paper include: proposing an automated knowledge graph construction method tailored for syllabi, integrating the education domain with deep learning techniques; designing a multi-stage entity-relationship extraction process to enhance knowledge extraction accuracy; constructing an evaluation dataset for the education domain to inform related research; and developing a complete system prototype that successfully validates the practicality and effectiveness of the proposed method. Flowchart as shown in Figure 1 and Figure 2.

2 Theoretical Section 1

This section primarily introduces the information extraction and analysis component.

In this study, entities are categorized into five types based on semantically significant vocabulary within the text: Course entities, including course names, codes, credit hours, etc.; Knowledge entities, including knowledge points, knowledge units, algorithms, data structures, etc.; Person entities, including authors, instructors, etc.; Tool entities, including programming languages, etc.; Other entities: including specialized terminology, concepts, etc.

This research formalizes the entity recognition task as sequence annotation. The objective is to find the optimal annotation sequence Y^* given an input sequence S , which can be formulated as the

following conditional probability maximization problem:

$$Y^* = \arg \max_Y P(Y|S) = \arg \max_Y \prod_{i=1}^n P(y_i | y_{1:i-1}, S) \quad (1)$$

Here, y_i denotes the annotation label for word w_i . $y_{1:i-1}$ represents the annotation history of the preceding $i-1$ words. As a sequential decision process, the prediction of the current label depends on the preceding label sequence and the input sentence.

To further model the joint probability $P(Y|S)$ of the annotation sequence Y , this study employs a linear chain conditional random field. This model treats the sequence annotation problem as a structured prediction task. The conditional probability distribution for structured prediction is defined as:

$$P(Y|S) = \frac{1}{Z(S)} \exp \left(\sum_{i=1}^n \sum_{k=1}^K \lambda_k f_k(y_{i-1}, y_i, S, i) \right) \quad (2)$$

Where $Z(S)$ is the normalization factor, f_k is the feature function, and λ_k is the weight corresponding to the feature function. When extracting keywords, duplicate instances must be avoided. For example, the terms “Beijing City” and “Beijing” share identical meanings. If both appear in the text, redundant “Beijing” should be removed while retaining “Beijing City” to minimize bloat in the knowledge graph. This study employs an entity similarity function to quantify the similarity between entity strings. This custom composite metric combines two distinct similarity measurement approaches.

$$\text{sim}(e, e') = \frac{v_e \cdot v_{e'}}{v_e v_{e'}} + \alpha \cdot \text{edit}(e, e') \quad (3)$$

Where v_e denotes the semantic vector representation of entity e , edit e, e' represents the edit distance, and α is the balancing parameter. This approach achieves the goal of measuring the similarity between two entities in semantic space. The deduplication process can be formalized as:

$$\begin{aligned} E_{\text{unique}} &= \{e \in E_{\text{final}} | \forall e' \in \\ E_{\text{final}}, e \neq e' \vee \text{sim}(e, e') < \tau\} \end{aligned} \quad (4)$$

To enhance the precision of relationships between nodes in knowledge graphs, it is necessary to capture the semantic relationships implied by entities within their context. This study designs a context fusion module $\text{ATTN}(h_s, h_o, H_C)$ based on an attention mechanism^{[5][9]}.

Specifically, the subject entity is represented as h_s , and the object entity as h_o . The association strength of each contextual word h_t with the current entity can be expressed as:

$$\beta_t = v_a^\top \tanh(W_s h_s + W_o h_o + W_c h_t) \quad (5)$$

Among these, W_s , W_o , and W_c are learnable projection matrices, while v_a is the attention weight vector. The attention distribution for each word is $\alpha_t = \frac{\exp(\beta_t)}{\sum_{j=1}^T \exp(\beta_j)}$. Based on this distribution, a context-aware joint representation of entity pairs is constructed by performing a weighted sum over the context word representations:

$$\text{ATTN}(h_s, h_o, H_C) = \sum_{t=1}^T \alpha_t h_t \quad (6)$$

For instance, in the sentence “Beijing is the capital of China,” the term ‘capital’ serves as key evidence for the “capital- affiliated country” relationship and functions as an indicator within this entity. Consequently, the attention mechanism will increase the weight assigned to the word “capital.”

Accurately processing lengthy academic texts such as syllabi presents a challenging task. It requires partitioning the long document D into a sequence of contiguous segments $\{C_1, C_2, \dots, C_k\}$ that satisfy length constraints while preserving semantic coherence and integrity within each segment. To achieve optimal decisions, this study formalizes the long document segmentation problem as a combinatorial optimization problem, aiming to find the segmentation scheme with minimal semantic loss under given constraints^[7]. The mathematical model is established as follows:

$$\min_{C_1, \dots, C_k} \sum_{i=1}^k \text{cost}(C_i) \quad (7)$$

The $\text{cost}(C_i)$ represents the semantic integrity cost function, which quantifies the semantic coherence of partition C_i . Three types of constraints are imposed: - Integrity constraint: Ensures every unit in document D is assigned to a single partition, guaranteeing no information omission. - Mutual exclusion constraint: Ensures partitions do not overlap, preventing redundant processing. - Technical constraint: Ensures partition sizes remain manageable.

$$\begin{aligned} \text{Subject to } & \bigcup_{i=1}^k C_i = D, C_i \cap C_j \\ & = \emptyset \text{ for } i \neq j, |C_i| \leq L_{\max} \quad \forall i \end{aligned} \quad (8)$$

This segmentation model provides a rigorous mathematical framework for processing long academic texts such as syllabi, thereby enhancing the accuracy of knowledge graph construction.

3 Theoretical Section 2

This section primarily introduces the construction aspects of knowledge graphs.

The construction of knowledge graphs requires evaluation criteria to optimize their accuracy and expertise. The evaluation methods adopted in this study fall into two major categories: set similarity-based evaluation and ranking quality-based evaluation. Set similarity-based evaluation treats knowledge graph construction as a process of set generation and matching, focusing on recall and precision. Ranking quality-based evaluation emphasizes the order in which the knowledge graph returns and organizes knowledge. Therefore, Mean Average Precision (mAP) is introduced to assess the system’s ability to rank relevant knowledge results.

$$\text{mAP} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{m_q} \sum_{k=1}^{m_q} P@k(q) \cdot \text{rel}_q(k) \quad (9)$$

Here, $P@k(q)$ denotes the precision rate for the top k results of query q , while $\text{rel}_q(k)$ represents the relevance of the k th result. This combined dynamic and static evaluation theory provides precise optimization directions for the knowledge graph construction process. To achieve the desired accuracy, the number of training iterations for knowledge graph samples must be evaluated. The assessment criterion requires that the probability of the generalization error $L(f)$ not exceeding the target error ϵ should be at least $1 - \delta$, necessitating n samples.

$$n \geq \frac{C}{\epsilon^2} \left(\text{VC}(\mathcal{H}) \ln \frac{1}{\epsilon} + \ln \frac{1}{\delta} \right) \quad (10)$$

Here, $\text{VC}(\mathcal{H})$ denotes the hypothesis space’s VC dimension, a classical measure of space complexity; ϵ represents the target error; and δ denotes the confidence level.

4 Experimental Section

4.1 Experimental Objective

This experiment aims to validate the effectiveness of the knowledge graph construction method for large language model-based syllabi. It evaluates the accuracy and recall rates of syllabus documents under entity extraction algorithms, verifies the capability of relation extraction algorithms to identify semantic relationships between knowledge points, and compares extraction performance differences across different courses. Three types of syllabi serve as the dataset: Data Structures, Object-Oriented Programming, and Compiler Principles. These syllabus documents contain structured information including course fundamentals, learning objectives, knowledge units, teaching content, and assessment requirements.

4.2 Experimental Environment

The system utilizes the Flask Web framework for development. Neo4j stores the knowledge graph, SQLite manages user accounts, and the large language model employs DeepSeek Chat API.

4.3 Experimental Process

Entity Extraction Algorithm:

The docx document is first preprocessed into chunks of approximately 1000–2000 characters each. Each chunk undergoes two independent rounds of entity extraction. After deduping, all chunk

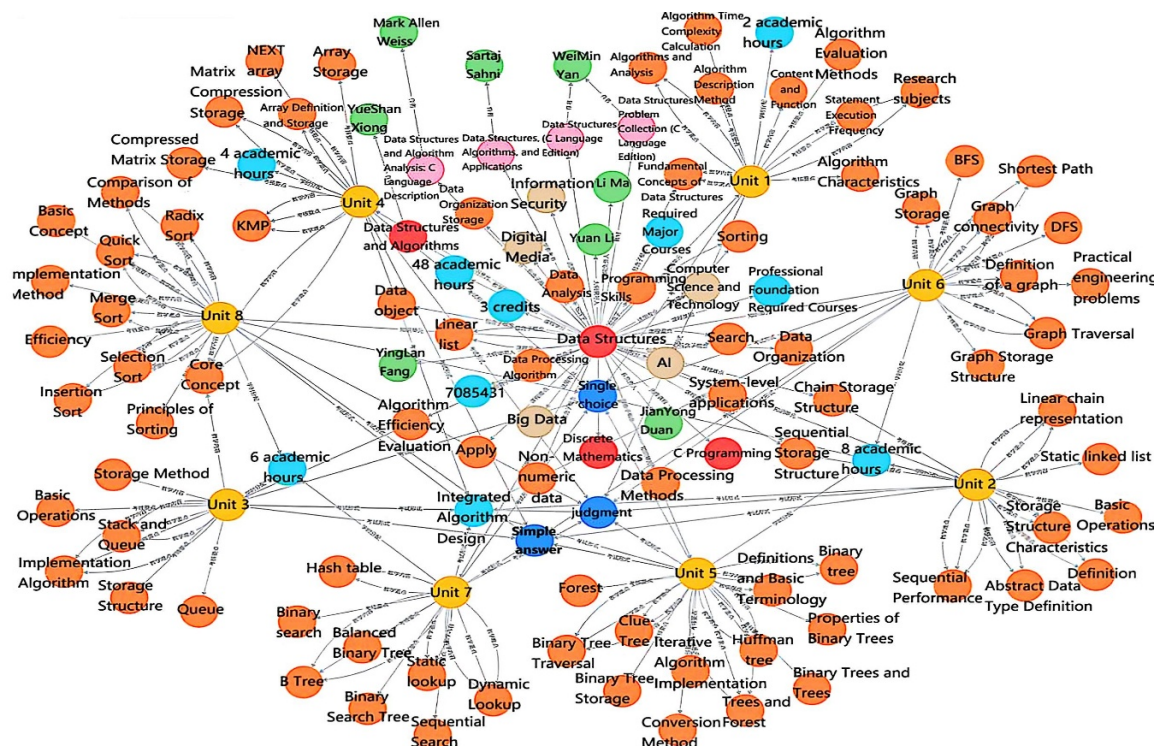


Figure 3: Data Structures Course Syllabus Knowledge Graph Overview Diagram

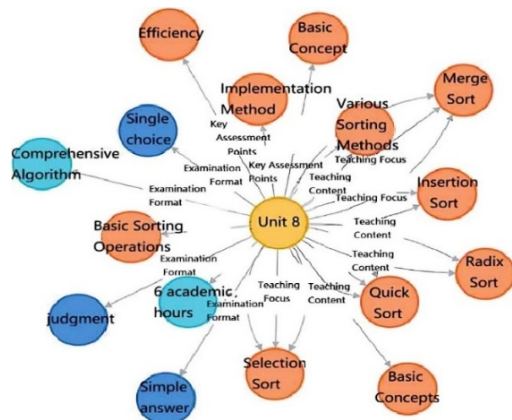


Figure 4: Knowledge Graph Overview Diagram Using Data Structures Unit 8 as an Example

results are merged and uniformly annotated with entity types. Irrelevant entities are filtered out, retaining only course-related terminology. The knowledge map of the data structures syllabus is shown in Figure 3.

Relationship Extraction Algorithm: Design corresponding prompt templates for different segmentation patterns, execute multi-round extraction, and cross-validate results^{[4][6]}. Finally,

eliminate duplicate relationships and validate their validity and consistency. An overview diagram of the knowledge graph is shown in Figure 4.

As shown in Table 1 and Table 2, the system achieved satisfactory entity extraction results across all three course syllabi, with the F1-value reaching 0.799. The data structures syllabus yielded the best extraction performance, attributed to its relatively well-defined concepts and more standardized terminology.

Overall performance is slightly below entity extraction. Effectively identifies primary relationship types Implied relationships and cross-chapter relationships still require partial improvement.

4.4 Experimental Conclusions

This experiment validates the feasibility of constructing knowledge graphs from syllabi using large language models. The system successfully converts unstructured syllabus documents into structured knowledge graphs. The harmonic mean values for entity extraction and relation extraction accuracy and recall reached 0.799 and 0.704 respectively, meeting practical application requirements. It demonstrates robust performance across diverse computer science syllabi, enabling partial visualization and intelligent retrieval of domain-specific course knowledge with promising application prospects.

Table 1: Evaluation of Entity Extraction Performance

Course	Accuracy rate	Recall rate	F1-value	Extract entity count	Standard Entity Count
Data Structures	0.847	0.792	0.818	156	142
Object-Oriented Programming	0.823	0.776	0.799	134	128
Compiler Theory	0.801	0.758	0.779	148	135
Average	0.824	0.775	0.799	146	135

Table 2: Relationship Extraction Performance Evaluation

Course	Accuracy rate	Recall rate	F1-value	Extract entity count	Standard Entity Count
Data Structures	0.756	0.683	0.717	89	76
Object-Oriented Programming	0.742	0.671	0.705	82	73
Compiler Theory	0.728	0.654	0.689	85	71
Average	0.742	0.669	0.704	85	73

5 Discussion Section

This study designed and implemented a deep learning-based curriculum syllabus knowledge graph construction system. It successfully transformed unstructured syllabus content into structured knowledge graphs.

Entity extraction accuracy consistently exceeded recall rates, indicating strong precision in entity recognition while leaving room for improvement in comprehensive entity coverage. Overall performance in relation extraction was slightly lower than entity extraction, particularly when processing cross-chapter content. This is attributed to limitations in current attention-based context modeling when handling semantic relationships.

By coordinating modules, the system enhances the accuracy and consistency of knowledge extraction. The system exhibiting potential for domain transfer and offering a viable solution for intelligent management of teaching resources. It holds significant application potential in areas such as curriculum visualization, personalized learning path recommendations, and knowledge gap analysis.

While the system yields satisfactory results, limitations remain. Knowledge extraction shows limited capability in recognizing abstract concepts, as evidenced by the relatively low F1 score for the “Compiler Principles” section. Recall rates for relation extraction decline when relation expressions are ambiguous or scattered across different paragraphs. The current system is sensitive to the quality and domain coverage of pre-training data, resulting in suboptimal graph construction when switching domains.

My recommendation is to explore domain adaptation techniques in future work. Fine-tuning the pre-trained model using academic papers or relevant textbooks from the education domain could enhance its precision in understanding entities and relationships, mitigating sensitivity to data quality. Intelligent educational applications could be developed based on the constructed knowledge graph.

6 Conclusion Analysis

This study successfully validated the feasibility of automatically constructing syllabus knowledge graphs using large language models. Experimental results demonstrate that the system has achieved practical application capability in entity extraction and relation extraction tasks compared to traditional inefficient manual construction methods, proving that natural language processing technology can effectively handle the task of extracting complex knowledge structures from semi-structured educational documents. The research architecture comprises: an information extraction model based on attention mechanisms and related theories; a long-document segmentation strategy grounded in optimization theory; and a comprehensive evaluation system. While room for improvement remains in areas like abstract concept recognition, at this stage, there are opportunities for application in areas such as course visualization, personalized learning path recommendations, and teaching effectiveness evaluation.

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